GeoTime

Multi-temporal satellite change analysis report

Location	Delhi, India
Period	2026-10-01 to 2026-10-31 (31 days)
Area	117.6 km2 (7.4 x 15.9 km)
Generated	31/10/2026, 12:08:12 am

Key metrics (area-wide, cloud-masked)

Metric	Earlier window	Later window	Change
Mean NDVI	-	-	-
Mean NDWI	-	-	-
Mean NDBI	-	-	-
NDVI above 0.4	-	-	-
Water (NDWI above 0.15)	-	-	-
Built-up (NDBI above 0.1)	-	-	-
Valid-data coverage (NDVI)	-	-	-
Fire hotspot detections	0	0	+0

Key findings

- The period is shorter than one year, so the trend statistics are confounded by seasonality and should not be read as long-term change.
- No usable NDVI data was found in one or both comparison windows.
- Pixel-level NDVI change (4,82,304 pixels cloud-free on both dates): 9.2% moved to a higher vegetation class and 6.5% to a lower class (net +2.7 pp). Strong shifts of two or more classes: 0.1% up, 0.1% down.
- Pixel-level surface water: 0.01 km2 gained and 0.26 km2 lost (net -0.25 km2); 0.26 km2 unchanged.
- Pixel-level built-up surface: 1.58 km2 gained and 2.59 km2 lost (net -1.01 km2); 6.10 km2 unchanged.
- True-colour comparison: 3.6% of pixels changed noticeably (RGB difference above 10%) and 0.4% strongly (above 20%). Spatial correlation of brightness between dates is 0.96; low values mean the scene layout changed. This layer includes clouds, so cross-check with the index layers.
- Fire hotspots (10-day windows): 0 detections earlier versus 0 later (+0); total fire radiative power 0.0 MW versus 0.0 MW. Hotspots include agricultural burning and industrial heat sources, not only wildfires.

1. Study area, data and parameters

Parameter	Value
Bounding box (W, S, E, N)	77.1712, 28.5418, 77.2468, 28.6859
Centre (lon, lat)	77.2090, 28.6139
Extent	7.38 x 15.93 km = 117.58 km2
Sensor / product	Copernicus Sentinel-2 L2A (surface reflectance)
Statistics grid	400 x 400 cells, about 18 m per cell
Change-map grid	471 x 1024 px, about 16 m per pixel
Earlier comparison window	2026-09-21 to 2026-10-11
Later comparison window	2026-10-21 to 2026-10-30
Time-series interval	5 days
Season offset between dates	30 days
Cloud handling	Scene Classification Layer: classes 4-7 kept; shadow, cloud, cirrus, snow removed
Fire data	NASA FIRMS VIIRS S-NPP (VIIRS_SNPP_NRT), 10-day windows

All areas are computed from pixel counts scaled to the study-area extent. Percentages refer to pixels with valid (cloud-free) data in the given window unless stated otherwise.

2. Time-series analysis

Area statistics for each 5-day interval over the selected period. Dots are the area mean, vertical bars are the spatial standard deviation within the area, and the dashed line is the Theil-Sen trend. Intervals with under 20% valid pixels are omitted.

NDVI - vegetation vigour

bars: spatial SD, dashed: Theil-Sen

No usable observations (too much cloud or no coverage).

NDWI - surface water and moisture

bars: spatial SD, dashed: Theil-Sen

No usable observations (too much cloud or no coverage).

NDBI - built-up and bare surfaces

bars: spatial SD, dashed: Theil-Sen

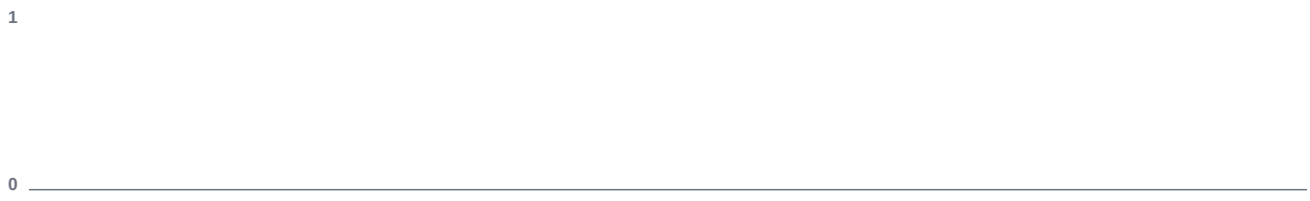
No usable observations (too much cloud or no coverage).

Trend and variability statistics

Index	n	Mean	SD	Minimum (date)	Maximum (date)	MK p	Sen/yr	R2
NDVI	0	needs 4+						
NDWI	0	needs 4+						
NDBI	0	needs 4+						

MK p is the Mann-Kendall trend-test p-value (below 0.05 suggests a monotonic trend). Sen/yr is the Theil-Sen slope in index units per year. Over periods shorter than a year the seasonal cycle dominates these values.

Valid-pixel coverage per interval (%, NDVI)



3. Earlier vs later window comparison

Distributions of each index over all valid pixels in the earlier (2026-09-21 to 2026-10-11) and later (2026-10-21 to 2026-10-30) windows. Red dashed lines mark the thresholds used in this report.

NDVI distribution

Earlier Later

Histogram unavailable.

Insufficient data for this index.

NDWI distribution

Earlier Later

Histogram unavailable.

Insufficient data for this index.

NDBI distribution

Earlier Later

Histogram unavailable.

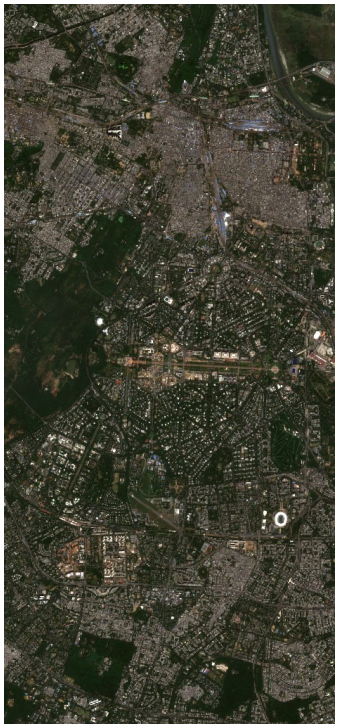
Insufficient data for this index.

4. Pixel-level change detection

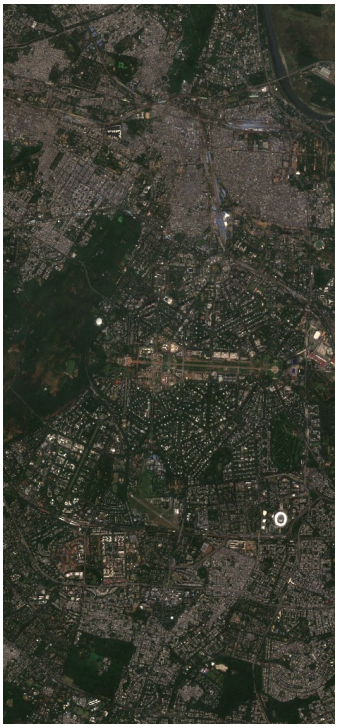
Change maps compare the earlier and later mosaics pixel by pixel, using only pixels that are valid on both dates.

4.1 True-colour difference (all-surface change)

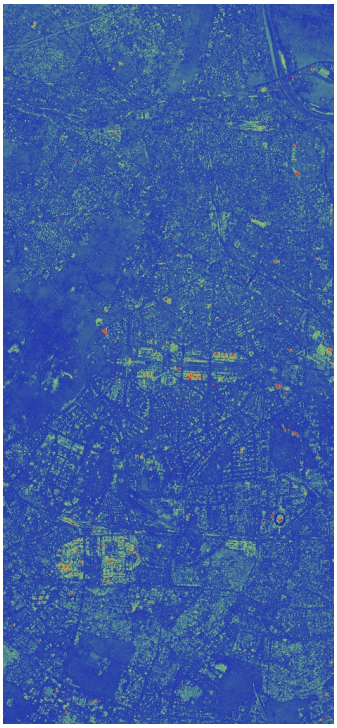
Earlier (2026-10-01)



Later (2026-10-31)



Change map



Pixels with valid data on both dates: 4,82,304 of 4,82,304 (100%). Pixels masked as cloud, shadow or no data on either date are excluded.

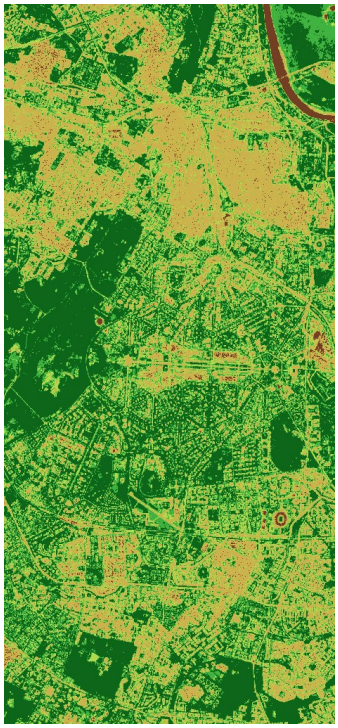
- Little or no change
- Moderate change
- Strong change (RGB difference of 30% or more)

Measure	Value
Mean RGB difference	3.7%
Pixels changed above 10%	3.6%
Pixels changed above 20%	0.4%
Brightness correlation between dates	0.958
Mean R, G, B earlier	62, 63, 48
Mean R, G, B later	58, 59, 49

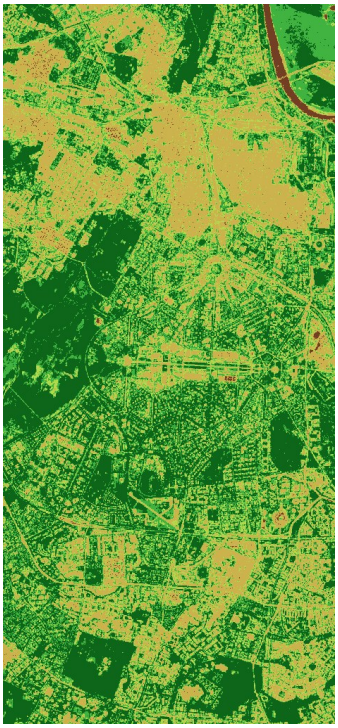
This layer is not cloud-masked, so clouds and haze appear as change. Use the NDVI, water and built-up pages to separate surface change from atmosphere.

4.2 Vegetation change and class transitions (NDVI)

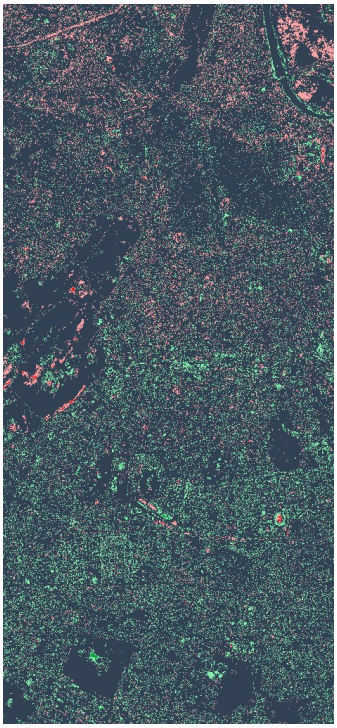
Earlier (2026-10-01)



Later (2026-10-31)



Change map



Pixels with valid data on both dates: 4,82,304 of 4,82,304 (100%). Pixels masked as cloud, shadow or no data on either date are excluded.

- Improved by 2 or more classes
- Improved by 1 class
- Same class
- Declined by 1 class
- Declined by 2 or more classes
- No valid data on one or both dates

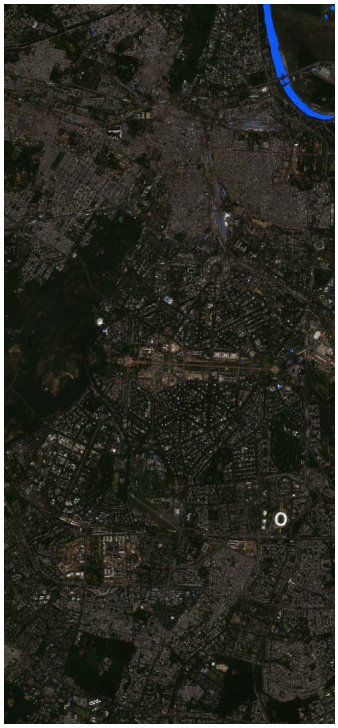
Change category	Share of valid pixels	Area km2
Improved (any)	9.2%	10.80
of which 2+ classes	0.1%	0.07
Unchanged	84.3%	99.12
Declined (any)	6.5%	7.66
of which 2+ classes	0.1%	0.10
Net balance	+2.7 pp	+3.15

Class transition matrix (% of valid pixels; rows = earlier, columns = later)

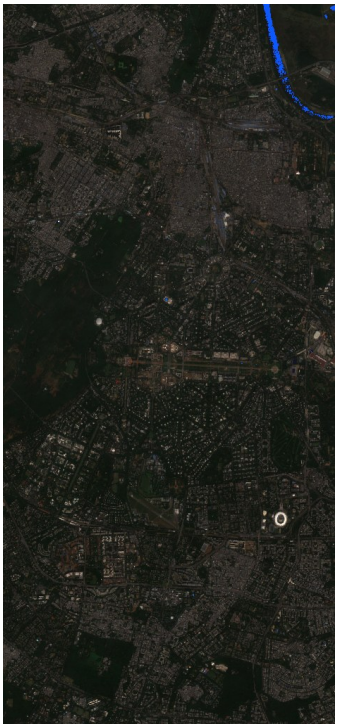
Earlier / Later	<0	0-0.2	0.2-0.4	0.4-0.6	>0.6
<0	0.5%	0.8%	0.0%	0.0%	0.0%
0-0.2	0.1%	22.2%	3.0%	0.0%	0.0%
0.2-0.4	0.0%	1.0%	13.8%	2.7%	0.0%
0.4-0.6	0.0%	0.0%	1.4%	17.1%	2.6%
>0.6	0.0%	0.0%	0.1%	3.9%	30.7%

4.3 Surface-water change (NDWI)

Earlier (2026-10-01)



Later (2026-10-31)



Change map



Pixels with valid data on both dates: 4,82,304 of 4,82,304 (100%). Pixels masked as cloud, shadow or no data on either date are excluded.

- water on both dates
- water gained (later only)
- water lost (earlier only)
- Other valid pixels
- No valid data on one or both dates

Category	Share of valid pixels	Area km2
Gained	0.0%	0.01
Lost	0.2%	0.26
Stable	0.2%	0.26
Net change	-0.21 pp	-0.25

4.4 Built-up surface change (NDBI)

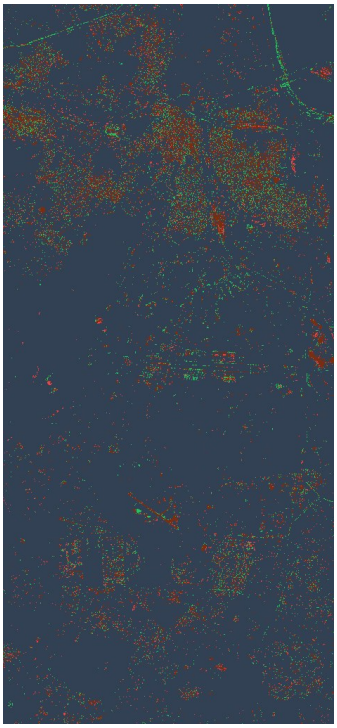
Earlier (2026-10-01)



Later (2026-10-31)



Change map



Pixels with valid data on both dates: 4,82,304 of 4,82,304 (100%). Pixels masked as cloud, shadow or no data on either date are excluded.

- built-up surface on both dates
- built-up surface gained (later only)
- built-up surface lost (earlier only)
- Other valid pixels
- No valid data on one or both dates

Category	Share of valid pixels	Area km2
Gained	1.3%	1.58
Lost	2.2%	2.59
Stable	5.2%	6.10
Net change	-0.86 pp	-1.01

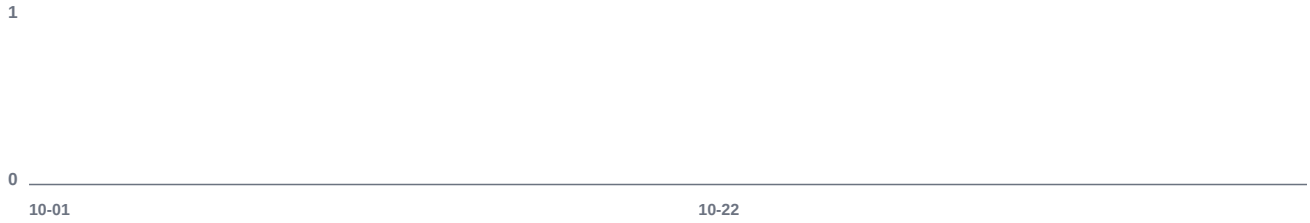
NDBI flags built-up surfaces but also dry bare soil and harvested fields. Treat gains as candidates for verification, especially when the dates are in different seasons.

5. Active fire hotspots (NASA FIRMS VIIRS)

Two equal 10-day windows are compared: 2026-10-01 to 2026-10-10 (earlier) and 2026-10-22 to 2026-10-31 (later). Data source: VIIRS_SNPP_NRT. Thermal anomalies include agricultural burning and industrial heat, not only wildfires.

Measure	Earlier	Later	Change
Detections	0	0	+0
High confidence	0	0	+0
Nominal confidence	0	0	+0
Low confidence	0	0	+0
Daytime detections	0	0	+0
Night-time detections	0	0	+0
Total FRP (MW)	0.0	0.0	+0.0
Mean FRP (MW)	-	-	-
Median FRP (MW)	-	-	-
Maximum FRP (MW)	-	-	-
Share in top-3 grid cells	0%	0%	-

Detections per day (blue: earlier window, orange: later window)



Spatial density (8 x 8 grid, north up)

Earlier **Later**

Densest cells (count at lat, lon). Earlier: none. Later: none.

6. Methods, interpretation guide and limitations

Data

Sentinel-2 L2A surface reflectance from the Copernicus Data Space Ecosystem via Sentinel Hub. Statistics use a 400 x 400 grid over the study area; change maps use the image grid listed in section 1. Mosaics prefer the scene with the lowest cloud cover.

Cloud masking

Time-series statistics and the NDVI, water and built-up change maps keep Scene Classification Layer classes vegetation, bare soil, water and unclassified, and remove cloud shadow, medium and high cloud, cirrus and snow. The true-colour layer is not masked. The SCL is itself imperfect; thin cloud and haze can survive.

Indices

NDVI = (B08 - B04) / (B08 + B04). NDWI = (B03 - B08) / (B03 + B08), water threshold 0.15. NDBI = (B11 - B08) / (B11 + B08), built-up threshold 0.1.

Effect size and distribution distance

Cohen's d is the mean difference divided by the pooled standard deviation (below 0.2 negligible, 0.2-0.5 small, 0.5-0.8 medium, above 0.8 large). Total variation is half the summed absolute difference between the two histograms. Wasserstein-1 is the area between the two cumulative distributions, in index units. No p-value is given for window differences because neighbouring pixels are spatially autocorrelated, which makes naive tests overconfident.

Trend tests

The Mann-Kendall test and Theil-Sen slope are non-parametric and robust to outliers, but they ignore autocorrelation and the seasonal cycle. Use series longer than one year, or deseasonalise before inferring long-term change.

Pixel-level change

Change maps compare the two rendered mosaics pixel by pixel and classify each pixel from its colour class, so they are quantised (NDVI in 5 classes, water and built-up as binary). Small misregistration, sun-angle and atmospheric differences can create spurious change at field edges.

Limitations

Comparison windows are about +/-10 days around each date, and the change-map mosaics may use up to +/-30 days when clear imagery is scarce, so they represent the window rather than one acquisition day. Fire hotspots are 375 m VIIRS thermal detections. Results are indicative and should be validated with ground data or high-resolution imagery before use in decisions or publication.

Appendix: time-series data

Interval	NDVI mean	NDVI SD	NDWI mean	NDBI mean	Valid

The full dataset (percentiles included) is saved alongside this report as a CSV file.